Function notation

1 $f(7) = -6$

2

a 112

b 966

3

a -20

b 4

4

a 12

b -16

5

a 8

b -13

6

a $\frac{35}{4}$

b -7

7

a 68

b -4

8 $\frac{9}{2}$

9

a 3

b $\sqrt{19}$

c $\sqrt{4}$

10

a 0

b 3

c $\frac{91}{25}$

11

a $a^2 + 8a$

b $b^2 + 8b$

12

a -4

b $-\frac{121}{32}$

13

a 0

b 2.67

c 80.99

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14

a $\sqrt{128}$ **b** 12 **c** $\sqrt{63}$ **d** $\sqrt{142}$

15

a 12 **b** $m^2 + 8$

16

a $y = 2 - \frac{x}{3}$ **b** $f(x) = 2 - \frac{x}{3}$ **c** $f(3) = 1$

17

a $y = \frac{x}{4} - 2$ **b** $f(x) = \frac{x}{4} - 2$ **c** $f(12) = 1$

18

a $y = 3 - x - 6x^2$ **b** $f(x) = 3 - x - 6x^2$

c -54

19

a $y = 4x^2 + x + 5$ **b** $f(x) = 4x^2 + x + 5$

c 23

20

a $y = \frac{6}{5}x + \frac{7}{5}$ **b** $f(x) = \frac{6}{5}x + \frac{7}{5}$ **c** 5

21

a 2 **b** 0 **c** $x = 1$

22

a 1

b Yes, because every value of x corresponds to only one value of y.

23

a

- Independent variable: Size of pizza
- Dependent variable: Cost of pizza

b

- Independent variable: Time spent studying
- Dependent variable: Mark achieved on a test

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c

- Independent variable: Amount borrowed
- Dependent variable: Duration of a loan

d

- Independent variable: Time spent exercising
- Dependent variable: Physical fitness level

24

a

i -48 **ii** 15 **iii** -49

b $x = 7.81, -7.81$

25

a $ak^3 - 3k + 5$ **b** $-ak^3 + 3k + 5$ **c** No **d** No

Algebra of functions

26 -46

27

a $A(5) = 26$

b $Q(4) = 52$

c $A(3) + Q(2) = 32$

28

a 39

b -25

c $a^3 - 8a + 7$

d $b^3 - 8b + 7$

e $(a+b)^3 - 8(a+b) + 7$

f No

29 $a^2 + 2ah + h^2 + 5a + 5h$

30 $a^2 + 10a + 24$

31

a $a^2 - 7a + 8$

b $2ah + h^2 - 3h$

32

a $4a^2 + 20a + 27$

b $8ah + 4h^2 + 4h$

33

a 12 **b** 42 **c** 32 **d** 18

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34 12

35

a $-5x^3 + 3x^2 + 35x - 21$ **b** 36

36

a $8x + 2$ **b** 34 **c** $-2x - 12$ **d** -32

Applications

37 $y = -7, -5$

38

a 320 **b** $x = -4, 2, -2$

39

a $c = -7$ **b** $a = b - 1$ **c** $21 = 4a + b$ **d** $a = 4, b = 5$

40

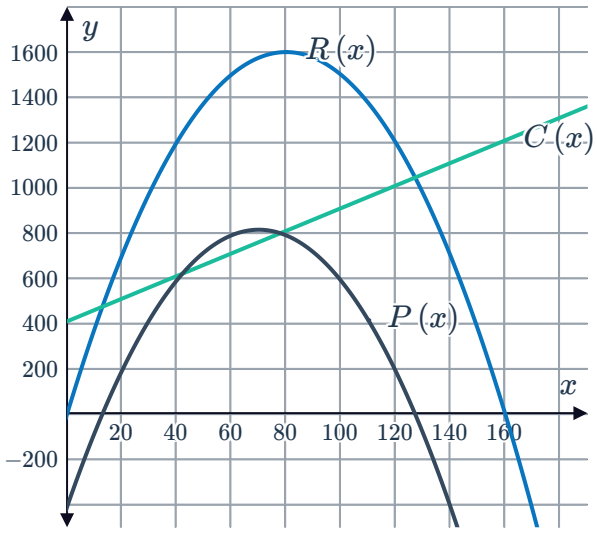
a $P(x) = -\dfrac{x^2}{4} + 35x - 410$

b

i $R(70) = 1575$ **ii** $C(70) = 760$

iii $P(70) = 815$

c



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41

a $T(x) = x^2 - 182x + 25\,650$

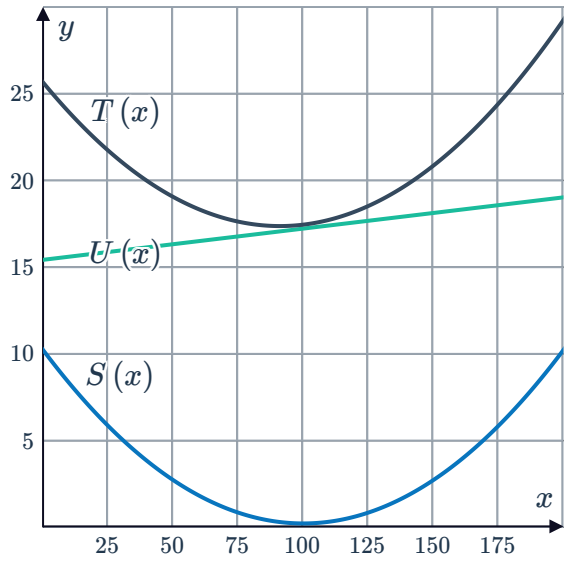
b

i $S(91) = 308$

ii $U(91) = 17\,061$

iii $T(91) = 17\,369$

c



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